

ROSWELL, NM, USA

WMO: 722680

Lat: 33.308N

Lon: 104.508W

Elev: 1112

StdP: 88.66

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23009

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.0	-5.8	-19.0	0.8	6.5	-16.9	1.0	7.4	13.9	14.8	12.0	13.2	2.1	0	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	38.8	17.9	37.3	18.1	36.0	18.2	21.5	30.4	20.9	30.0	20.5	29.6	5.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.3	23.4	18.8	15.6	23.1	18.2	15.0	23.1	67.9	30.3	65.8	29.9	63.9	29.6	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.5	8.7	-12.4	41.4	3.4	1.3	-14.9	42.3	-16.9	43.0	-18.8	43.7	-21.3	44.6		
			-13.4	23.3	3.2	2.2	-15.7	24.9	-17.5	26.1	-19.3	27.3	-21.6	28.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	5.4	8.1	12.4	16.7	21.8	26.6	27.7	26.9	23.1	16.8	10.3	5.2
	DBStd	8.98	4.57	4.92	4.60	4.23	3.86	2.60	2.20	2.30	3.16	4.18	4.41	4.74
	HDD10.0	484	152	83	30	6	0	0	0	0	0	5	49	158
	HDD18.3	1695	401	287	187	76	14	0	0	0	5	76	243	406
	CDD10.0	2964	9	31	104	207	366	499	548	522	392	217	57	11
	CDD18.3	1134	0	0	4	28	122	249	289	264	147	30	1	0
	CDH23.3	14712	5	39	212	650	1764	3334	3575	3034	1564	482	51	3
	CDH26.7	7828	0	6	49	239	912	1982	2048	1662	751	174	6	0
Wind	WSAvg	3.8	3.3	3.9	4.3	4.8	4.5	4.5	3.7	3.4	3.5	3.5	3.3	3.2
Precipitation	PrecAvg	314	10	11	12	13	25	39	48	53	44	29	14	15
	PrecMax	630	38	51	72	73	116	128	175	165	167	150	75	78
	PrecMin	74	0	0	0	0	0	0	0	2	0	0	0	0
	PrecStd	120	9	12	15	18	28	35	45	40	40	37	17	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	27.3	30.5	33.8	38.0	40.9	40.2	39.4	36.7	32.9	27.5	23.4
		MCWB	9.9	11.2	12.1	13.4	15.7	17.1	18.9	19.0	18.9	15.8	12.1	9.9
	2%	DB	20.8	24.3	28.0	31.3	36.0	39.3	38.1	37.2	34.8	30.7	24.7	20.2
		MCWB	8.7	10.1	11.4	12.6	15.2	17.1	19.0	19.3	18.0	15.3	11.6	8.8
	5%	DB	17.9	21.6	25.8	29.5	34.2	37.7	36.7	35.8	33.2	28.6	22.2	17.3
		MCWB	7.3	8.8	10.7	12.4	14.8	17.2	19.1	19.3	17.8	14.7	10.7	7.5
	10%	DB	14.9	18.5	23.2	27.5	32.0	35.7	35.3	34.3	31.3	26.0	19.5	14.3
		MCWB	6.0	7.6	9.8	11.7	14.5	17.3	19.1	19.4	17.8	14.3	9.6	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	12.2	14.9	16.3	19.5	22.2	22.5	21.9	21.7	20.2	15.2	11.2
		MCDB	21.3	24.2	23.7	24.9	28.8	31.4	31.2	30.6	29.7	25.6	20.6	19.1
	2%	WB	9.3	10.8	12.9	15.1	18.4	20.8	21.5	21.3	20.6	18.7	13.3	9.7
		MCDB	18.7	22.1	23.6	24.2	28.2	30.7	30.7	30.2	28.1	23.5	20.5	18.7
	5%	WB	7.9	9.5	11.6	14.0	17.4	20.1	20.9	20.8	19.9	17.0	11.7	8.1
		MCDB	16.7	20.4	22.7	24.8	27.7	30.1	30.3	29.8	27.1	23.6	20.2	16.5
	10%	WB	6.5	8.1	10.4	12.8	16.4	19.4	20.4	20.4	19.2	15.6	10.4	6.6
		MCDB	14.6	17.6	21.4	24.5	26.8	29.4	29.9	29.5	26.5	23.0	18.3	13.8
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.7	17.2	17.5	16.9	16.5	14.3	14.0	14.3	15.4	15.9	15.2
		MCDBR	21.5	22.1	21.7	20.8	20.3	19.7	16.6	16.1	17.4	19.4	20.3	20.3
		MCWBR	11.8	11.2	10.1	9.0	8.2	6.8	4.9	4.6	5.9	8.1	10.2	11.1
	5% WB	MCDBR	20.0	20.7	19.4	18.0	16.9	15.7	13.7	13.6	13.4	14.4	17.5	18.4
		MCWBR	11.3	10.8	9.5	8.6	7.4	5.5	4.2	3.9	5.0	6.8	9.3	10.4
Clear-Sky Solar Irradiance	taub		0.250	0.259	0.278	0.308	0.327	0.354	0.384	0.364	0.335	0.292	0.268	0.256
	taud		2.615	2.596	2.535	2.432	2.436	2.400	2.364	2.412	2.451	2.576	2.576	2.591
	Ebn at Noon		991	1011	1006	979	956	926	895	912	929	957	957	962
	Edh at Noon		75	85	98	114	115	119	123	115	106	86	77	73
All-Sky Solar Radiation	RadAvg		3.46	4.42	5.67	6.99	7.58	7.75	7.27	6.63	5.72	4.62	3.64	3.02
	RadStd		0.29	0.38	0.37	0.42	0.41	0.35	0.35	0.26	0.36	0.43	0.30	0.24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	+136	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page